SAURABH PATEL

Electronics & Communication Engineering | Embedded Systems | Robotics

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LinkedIn | GitHub | Portfolio

Professional Summary

Electronics and Communication Engineering student at NIT Nagaland with hands-on project experience in embedded systems, sensor integration, wireless communication, robotics, UAVs, and AI-assisted screening. Interested in embedded hardware, device-level development, prototyping, and electronics engineering internships.

Education

National Institute of Technology Nagaland Aug 2024 – Jun 2028
B.Tech. in Electronics and Communication Engineering

St. Xavier School, Bhiwadi 2021 – 2022
Class XII – Physics, Chemistry and Mathematics

St. Xavier School, Bhiwadi 2019 – 2020
Class X

Technical Skills

- **Programming:** C, Embedded C, MySQL, HTML, CSS, Python (Basic)
- **Microcontrollers & Platforms:** ESP32, STM32, Arduino, Raspberry Pi
- **Embedded Systems:** Sensor interfacing, GPIO, PWM, ADC, timers & interrupts, GPS integration, motor control, hardware debugging, circuit prototyping, soldering
- **Communication:** UART, I^2C, SPI, Wi-Fi, Bluetooth, ESP-NOW
- **Robotics & UAV:** Multirotor assembly, flight controllers, IMU integration, obstacle avoidance, ultrasonic sensing, ESC/BLDC motor control
- **AI & Vision:** Edge Impulse, image classification, gait/video analysis
- **Tools:** Arduino IDE, STM32CubeIDE, MATLAB, Git, GitHub, VS Code
- **Hardware & Lab:** Multimeter, soldering station, PCB/circuit prototyping

Major Projects

MineSafe-X – Low-Visibility Mining Vehicle Safety System *ESP32, sensors, wireless communication*

- Developing a retrofit driver-assistance prototype for mining vehicles operating in fog and low visibility.
- Integrates multi-sensor distance monitoring, vehicle-to-vehicle communication, tracking, and collision-warning concepts.

Aerial Fire Detection & Suppression Drone *STM32, ESP32, Raspberry Pi, Edge Impulse*

- Developing an aerial prototype for AI-assisted fire recognition, GPS/IMU-based sensing, and payload deployment.
- Explores onboard image classification and embedded control.

Ostereo-AI – Osteoarthritis Risk Marker Screening System *AI, gait analysis*

- AI-assisted screening workflow combining questionnaires, walking-video/gait analysis, and knee X-ray inputs to support risk assessment and referral.

AC Line Fault Detection System

- Prototype concept for detecting and indicating AC line fault conditions through electronic sensing and indication.

Additional Projects: Multirotor UAV, ultrasonic obstacle detection, Bluetooth-controlled RC car.

Experience

Electronics Manufacturing Intern | Elexoplus India Private Limited, Bhiwadi, Rajasthan 21 June – 19 July 2026

- Gained practical exposure to electronics manufacturing, component assembly, production workflows, and system testing.

Sanchar Mitra | Department of Telecommunications (DoT) August 2025 – Present

- Supporting telecom awareness activities and public engagement initiatives.

Workshops & Training

Workshop on Assembly of Obstacle Avoidance Based on Multirotor UAV | NIELIT September 2025

- Hands-on workshop on assembling a multirotor UAV with an obstacle-avoidance system; certificate of participation received.

Achievements

- Runner-Up, Code For Nagaland 26-Hour Hackathon, Wildcard Track – Open Innovation, Team Ostereo-AI, October 2026.
- Invited Exhibitor, TXG Expo Nagaland, Aerial Recognition project, 2026.
- Winner, Robo GP, Technical Board, NIT Nagaland, March 2025.

Leadership & Positions of Responsibility

- Robotics Secretary, Robotics Club, NIT Nagaland August 2026 – Present
- Joint Secretary, Entrepreneurship & Incubation Cell, NIT Nagaland
- Internship Coordinator, NIT Nagaland August 2026 – Present
- Campus Ambassador, The Polaris Fellowship September 2026 – Present
- Assistant Club Secretary, Robotics August 2025 – August 2026

Languages

Hindi | English | Haryanvi